1.Answer each of the following:

a. Suppose that a simple regression has quantities $N = 20, \sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2.

$$R^2 = \frac{SSR}{SST}\ , SST = \sum (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2$$

$SST = 7825.94 - 20 * (19.21)^2 = 445.458$, $R^2 = 375.47/445.458 \approx 0.8429$

b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{SSE}{n-2}\ , SST = SSR + SSE\ , R^2 = \frac{SSR}{SST}$$

$SSR = R^2 * SST = 0.7911 * 725.94 = 574.291134$, $SSE = SST - SSR = 725.94 - 574.291134 = 151.648866$, $\hat{\sigma}^2 = \frac{151.648866}{20-2} \approx 8.4249$

c. Suppose that a simple regression has quantities $\sum(y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2.

$$R^2 = 1 - \frac{SSE}{SST}$$

$SST = \sum(y_i - \bar{y})^2 = 631.63$, $SSE = \sum \hat{e}_i^2 = 182.85$, $R^2 = 1 - \frac{182.85}{631.63} \approx 0.7105$